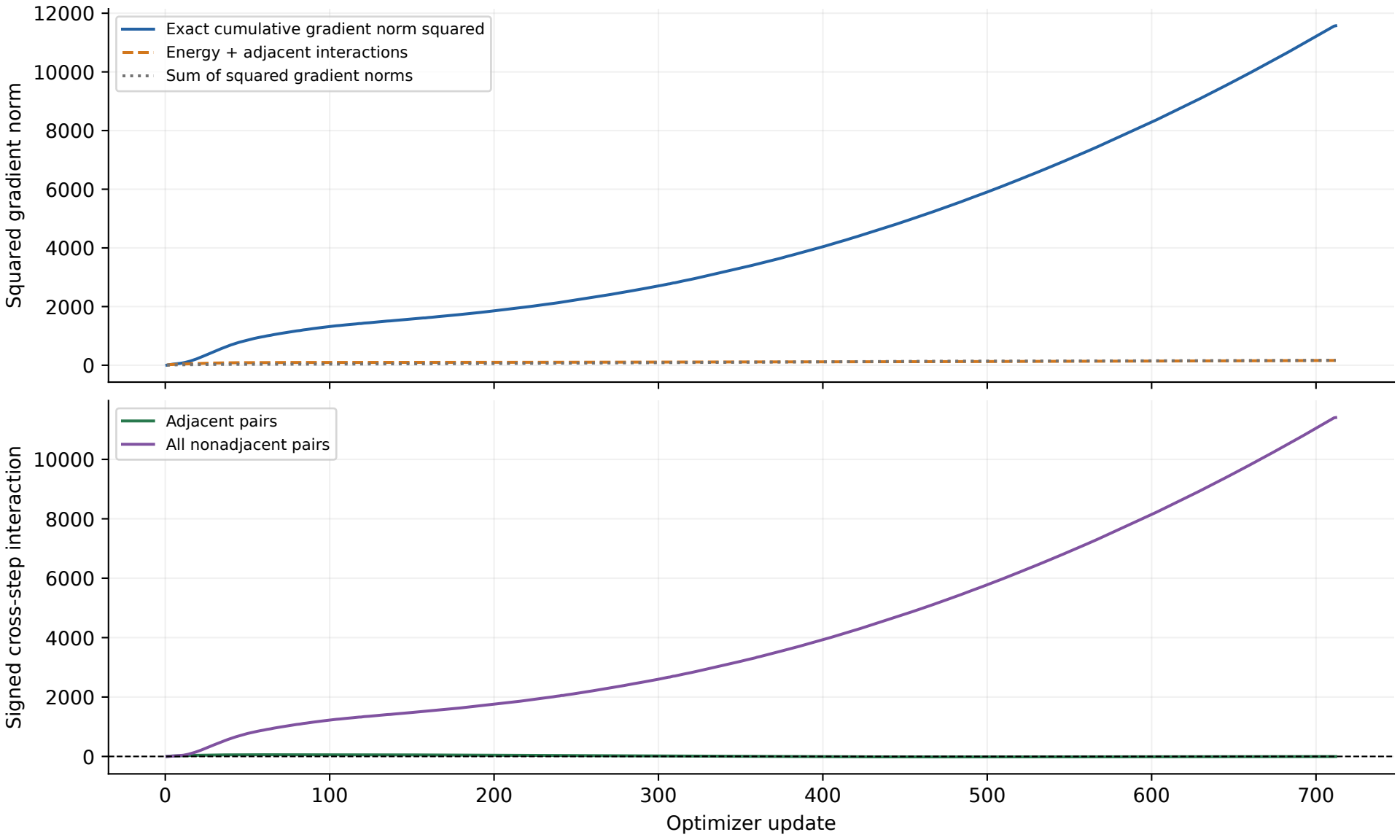
Exact cumulative gradient norm squared
Energy + adjacent interactions
Sum of squared gradient norms
Squared gradient norm
12000
10000
8000
6000
4000
2000
0
Adjacent pairs
All nonadjacent pairs
Signed cross-step interaction
10000
8000
6000
4000
2000
0
0
100
200
300
400
500
600
700
Optimizer update